benrcarver@gmail.com
github.com/scusemua
scusemua.github.io
571 535 7343

Benjamin Carver

Summary

- PhD candidate in computer science with extensive experience designing, developing, and evaluating large, scalable, high-performance distributed systems across multiple cloud platforms.
- Adaptive learner and adept problem-solver with proficiency in a wide array of programming languages, development tools, frameworks, and cloud services.
- Established, facilitated, and lead multiple research collaborations with leading academic and industry partners.
- Motivated and driven to leverage strong academic and development background, along with substantial research expertise in distributed systems, cloud and serverless computing, and AI/ML, to help organizations build innovative, scalable, and high-performance infrastructure, products, and services.

Education

George Mason University, Fairfax, Virginia

May 2025 *Computer Science*, Doctor of Philosophy, (GPA: 4.0).

May 2021 *Computer Science*, Master of Science, *Summa cum laude* (GPA: 4.0).

May 2020 *Computer Science*, Bachelor of Science, *Summa cum laude* (GPA: 4.0).

Employment

Spring 2019–Ongoing **Undergraduate/Graduate Research Assistant**, George Mason University.
- Served as the sole developer and tester of 3 complete, large-scale open-source distributed systems and as a co-developer of 2 other distributed applications of similar scale.
- Experience creating and deploying distributed applications and services in cloud environments–including AWS, Google Cloud Platform, Microsoft Azure, and IBM Cloud–using Kubernetes and Docker (Compose/Swarm).
- Published 4 academic papers in top-tier academic conferences, 3 of which were first-authorship publications, with 3 additional first-authorship papers under review or close to submission.
- Established, lead, and facilitated multiple collaborations with leading academic and industry research labs.

May–August 2022 **Microsoft Research Intern**, Microsoft Research Lab - Redmond.
- Team member responsible for designing, developing, and evaluating a new microsecond-latency FaaS platform.
- Improved API usability and reduced overheads by 25% by re-implementing client-facing interfaces as CPython extension modules, enabling multi-threading support and improving client application performance.

June 2016–July 2020 **Software Developer**, BRTRC Federal Solutions.
- Optimized a .NET-based Geographic Information Systems (GIS) application, reducing application load time by 5$\times$ and application memory footprint by 70%, significantly improving usability and the overall user-experience.
- Implemented system for manipulating and visualizing realistic, user-defined terrain maps and custom 3D models.

Ongoing Research

Jupyter Notebooks-as-a-Service (NaaS) Platform

- Responsible for the design, implementation, and evaluation of a novel, large-scale, fully functional Jupyter Notebooks-as-a-Service platform with full support for GPU-based workloads like deep learning training.
- Developed novel design that enables support for new classes of AI/ML applications and workloads.
- Enabled detailed, real-time metric and log collection, visualization, and monitoring by integrating platform with multiple industry-standard frameworks and technologies–including Prometheus, Promtail, Loki and Grafana.
- Created robust, detailed simulation software to evaluate pricing/cost, scheduling, fault tolerance, and scaling algorithms and policies of the Jupyter NaaS platform at extremely large scales.

A Microsecond-Scale Functions-as-a-Service (FaaS) Platform

- Team member responsible for designing, implementing, and evaluating a novel FaaS platform created to support modern scientific and data-analytics-based workloads.
- Integrated support for latest networking and virtualization technologies to enable orders-of-magnitude reduction in overhead incurred by the serverless platform.

Enhancing FaaS-Based Storage Systems for Privacy & Security

- Tasked with designing a secure FaaS-based storage system that incorporates techniques from cryptography, network security, and database systems to provide strong security guarantees.

Publications

[ASPLOS '23] **Benjamin Carver, Runzhou Han, Jingyuan Zhang, Mai Zheng, Yue Cheng**, *λFS: A Scalable and Elastic Distributed File System Metadata Service using Serverless Functions*, ACM International Conference on Architectural Support for Programming Languages and Operating Systems.
- Designed, implemented, and evaluated λFS, a first-of-its-kind serverless metadata service for large-scale distributed file systems like Hadoop Distributed File System (HDFS).
- Designed and implemented elastic auto-scaling mechanisms, efficient client libraries, and a novel FaaS-based metadata cache to provide 4.13× higher throughput, 90.40% lower latency, and 85.99% lower user-side cost compared to state-of-the-art baselines for real-world industrial workloads.

[VLDB '23] **Ao Wang, Jingyuan Zhang, Xiaolong Ma, Benjamin Carver, Ali Anwar, Lukas Rupprecht, Dimitrios Skourtis, Vasily Tarasov, Feng Yan, Yue Cheng**, *InfiniStore: Elastic Serverless Cloud Storage*, International Conference on Very Large Databases.
- Team member assisting in the testing and evaluation of InfiniStore, a first-of-its-kind serverless-function-based data storage service that achieves 97.24% user-side cost savings compared to AWS ElastiCache.
- Implemented a lightweight, microservice-based workload driver in Golang to execute large-scale, customizable MapReduce workloads as part of the evaluation of InfiniStore and its predecessor, InfiniCache.

[SOCC '20] **Benjamin Carver, Jingyuan Zhang, Ao Wang, Ali Anwar, Panruo Wu, Yue Cheng**, *Wukong: A Scalable and Locality-Enhanced Framework for Serverless Parallel Computing*, ACM Symposium on Cloud Computing (AR: 35/143 = 24.4%).
- Designed, implemented, and evaluated Wukong, a fully-featured serverless-computing-based execution engine created atop AWS Lambda and AWS Fargate.
- Developed novel scheduling algorithms that leverage serverless functions to execute large-scale parallel computing jobs up to 68.17× faster while reducing user-side cost by 92.96% compared to state-of-the-art baselines.

[PDSW '19] **Benjamin Carver, Jingyuan Zhang, Ao Wang, Yue Cheng**, *In Search of a Fast and Efficient Serverless DAG Engine*, International Parallel Data Systems Workshop.
- Created and evaluated system prototype that uses AWS Lambda to improve the performance of real-world data analytics, linear algebra, and machine learning workloads by up to 3.1× compared to a non-serverless baseline.

Skills

Languages: Python, Golang, Java, C/C++, C#, JavaScript/TypeScript, CSS.

Development Tools: VS Code, Visual Studio, JetBrains IDEs (IntelliJ/GoLand/WebStorm), Linux, Microsoft Office, Git (GitHub/GitLab/BitBucket), Jupyter Notebooks.

Frameworks & Services: Kubernetes, Docker, Docker Compose, Docker Swarm, Prometheus, Grafana, Loki, Promtail, HDFS, Dask, Pandas, NumPy, SciPy, PyTorch, JupyterLab, React, ZooKeeper, Redis, MySQL, Ansible.

Cloud: **Amazon Web Services (AWS)**
EC2, VPC, Lambda, S3, IAM, CloudWatch, Fargate, ECS, EKS.
Google Cloud Platform (GCP)
Compute Engine, Google Kubernetes Engine (GKE), Cloud Functions.
Microsoft Azure Cloud, **IBM Cloud**

Honors & Awards

August 2023 **VLDB2023 NSF Travel Grant**.

May 2021 **George Mason University Presidential Scholarship Award**.
Renewed annually for 4 years.

May 2021 **Distinguished Academic Achievement Award**.

May 2020 **Distinguished Undergraduate Research Award**.

May 2020 **Distinguished Academic Achievement Award**.

Relevant Academic Coursework

Systems: Operating Systems, Distributed Systems, Database Systems, Concurrent and Distributed Systems.

AI/ML: Large-Scale Optimization for Deep Learning, Deep Learning, Advanced Artificial Intelligence, Artificial Intelligence, Machine Learning, Data Mining.

Core: Data Structures, Computer Architecture, C Programming, Compilers, Object Oriented Software, Software Engineering, Algorithm Analysis, Graph Algorithms, Mobile App Development, Computer Networks.